From Last Time ...

We solved $A\vec{x} = \vec{b}$:

$$(A \mid \vec{b}) \longrightarrow \text{ref}(A \mid \vec{b}).$$

The same problem can be written as

$$x_1 \vec{a}_1 + x_2 \vec{a}_2 + \cdots + x_n \vec{a}_n = \vec{b},$$

where $\vec{a}_i$ is the ith column of A.

Linear Transformation

A function $T: X \to Y$ is a rule that associates each element $x \in X$ to a unique element $y \in Y$.

A function $T: \mathbb{R}^m \to \mathbb{R}^n$ is a linear transformation if

$$T(\vec{x}) = A\vec{x}$$

for some matrix $A_{n \times m}$. This matrix is the standard matrix of T.

Ex. Consider

$$T(\vec{x}) = \begin{pmatrix} 5 & 1 \\ 2 & -1 \end{pmatrix} \vec{x}.$$

The transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ takes $\vec{x} \in \mathbb{R}^2$ and sends it to

$$\begin{pmatrix} 5 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5x + y \\ 2x - y \end{pmatrix}.$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5x + y \\ 2x - y \end{pmatrix}$$

For example:

$$T\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 - 2 \\ 2 + 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Ex. The identity transformation is

$$T: \mathbb{R}^n \to \mathbb{R}^n, \qquad T(\vec{x}) = \vec{x}.$$

If $\vec{y} = T(\vec{x})$:

$$\begin{aligned} y_1 &= x_1 \\ y_2 &= x_2 \\ y_3 &= x_3 \\ &\vdots \end{aligned}$$

Ex. The identity transformation is

$$T : \mathbb{R}^n \to \mathbb{R}^n, \qquad T(\vec{x}) = \vec{x}.$$

If $\vec{y} = T(\vec{x})$:

$$\begin{aligned} y_1 &= x_1 \\ y_2 &= \phantom{x_1 +{}} x_2 \\ y_3 &= \phantom{x_1 + x_2 +{}} x_3 \\ &\vdots \\ y_n &= \phantom{x_1 + x_2 + x_3 + \cdots +{}} x_n \end{aligned}$$

$$\Rightarrow \vec{y} = x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \cdots + x_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

$$\Rightarrow \vec{y} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \vec{x}$$

This is the identity matrix!

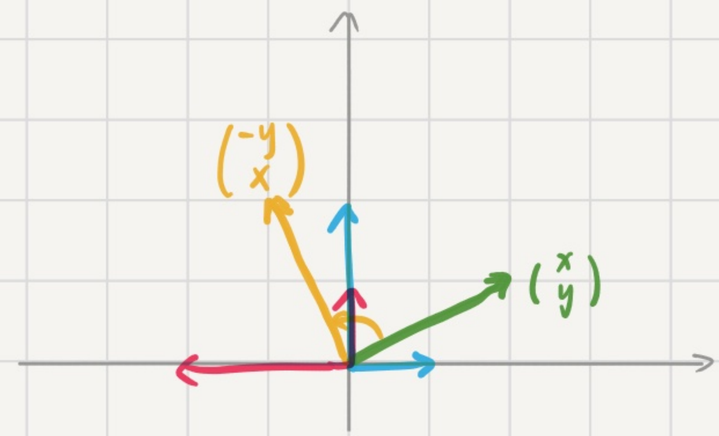
Ex. Consider

$$T(\vec{x}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \vec{x}.$$

What is:

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}.$$



This rotates "L" by 90° counterclockwise.

For an arbitrary vector:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

Theorem 1. Let $T: \mathbb{R}^m \to \mathbb{R}^n$ be a linear transformation. The standard matrix of T is given by

$$A = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_m) \\ \downarrow & \downarrow & & \downarrow \end{pmatrix},$$

where $\vec{e}_i$ is the ith standard basis vector.

Proof. Let $\vec{v}_i$ be the ith column of A. Then,

$$\vec{v}_i = A\vec{e}_i = T(\vec{e}_i).$$

$\square$

Note that every vector $\vec{x} \in \mathbb{R}^m$ can be written as a linear combination of the standard basis vectors:

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} + \cdots + x_m \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$$

$$\Rightarrow \vec{x} = x_1\vec{e}_1 + \cdots + x_m\vec{e}_m.$$

Theorem 2. $T: \mathbb{R}^m \to \mathbb{R}^n$ is a linear transformation if and only if:

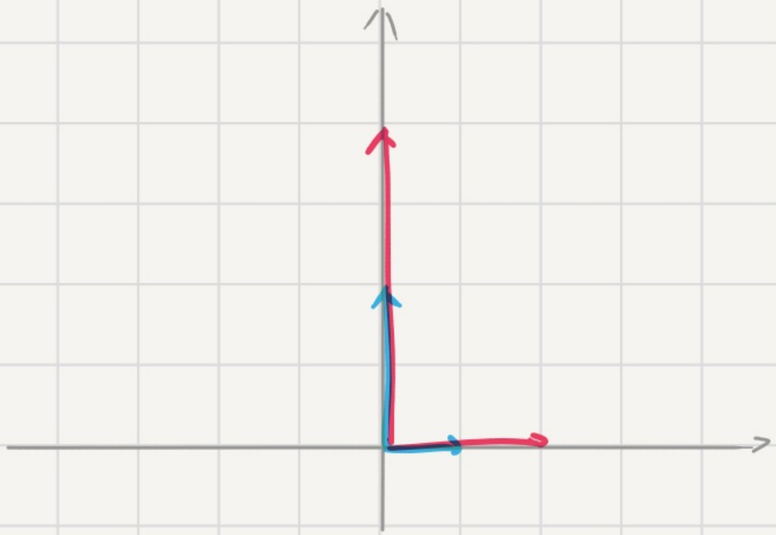
1. $T(\vec{v} + \vec{w}) = T(\vec{v}) + T(\vec{w})$, AND
2. $T(k\vec{v}) = kT(\vec{v})$ for $k \in \mathbb{R}$.

Geometry in $\mathbb{R}^2$

Ex. $T(\vec{x}) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\vec{x} = 2I\vec{x} = 2\vec{x}.$

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$



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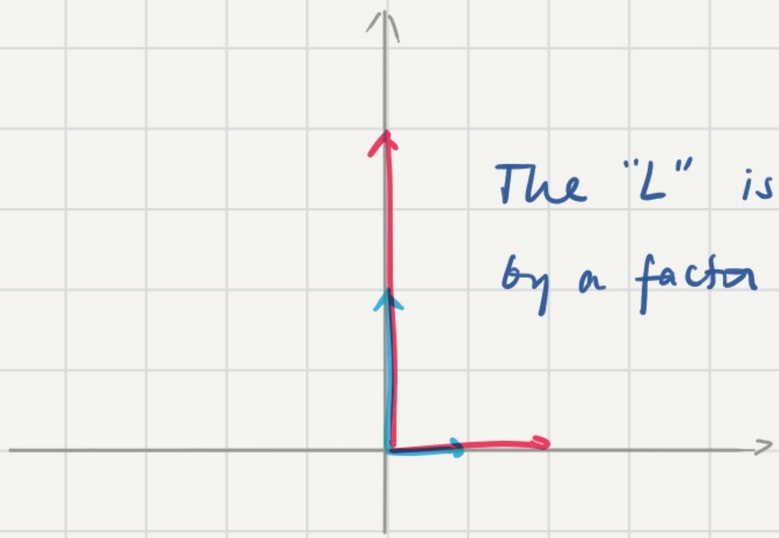


The "L" is enlarged by a factor of 2.

Scaling

Standard Matrix

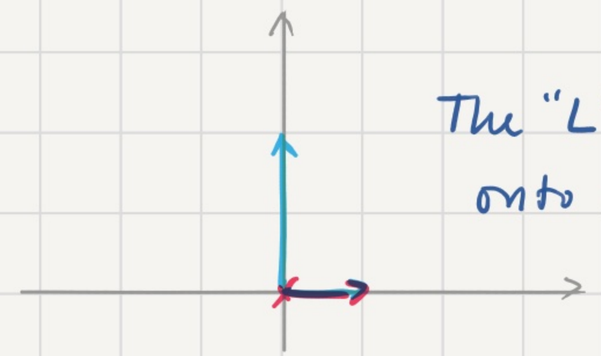
$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}, \quad k > 0$$

- $k > 1$: enlarges by k
- $0 < k < 1$: contract by k

Ex. $T(\vec{x}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \vec{x}$

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



The "L" is projected onto the x-axis.

Orthogonal Projections

Let $l = t\vec{v}$ be a line in $\mathbb{R}^2$.

Then, every vector $\vec{x} \in \mathbb{R}^2$ can be written as $\vec{x} = \vec{x}^{\parallel} + \vec{x}^{\perp}$, where $\vec{x}^{\parallel}$ is parallel to l, $\vec{x}^{\perp}$ is orthogonal to l.

Orthogonal Projections

Let $\ell = t\vec{v}$ be a line in $\mathbb{R}^2$.

Then, every vector $\vec{x} \in \mathbb{R}^2$ can be written as $\vec{x} = \vec{x}'' + \vec{x}^{\perp}$, where $\vec{x}''$ is parallel to ℓ, $\vec{x}^{\perp}$ is orthogonal to ℓ.

The orthogonal projection of $\vec{x}$ onto ℓ is

$$T(\vec{x}) = \vec{x}''.$$

We write $T(\vec{x}) = \text{proj}_{\ell}(\vec{x})$.

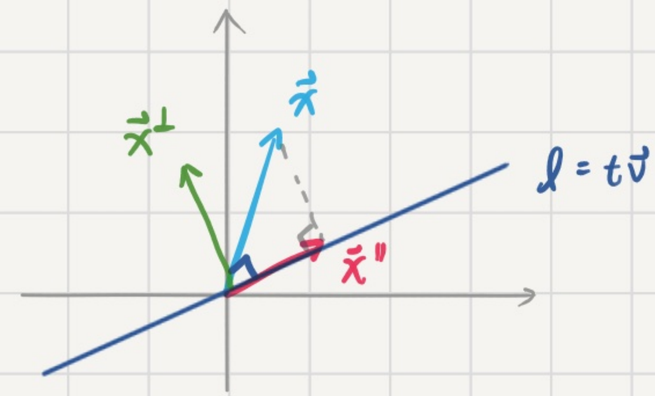
Standard Matrix

$$\text{proj}_{\ell}\vec{x} = \left(\frac{\vec{x}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\right)\vec{v} = \left(\frac{\overbrace{\begin{pmatrix} x \\ y \end{pmatrix}\cdot\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}}^{xv_1 + yv_2}}{v_1^2 + v_2^2}\right)\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$= \frac{1}{v_1^2+v_2^2}\begin{pmatrix} (v_1^2)x + (v_1v_2)y \\ (v_1v_2)x + (v_2^2)y \end{pmatrix}$$

$$= \frac{1}{v_1^2+v_2^2}\begin{pmatrix} v_1^2 & v_1v_2 \\ v_1v_2 & v_2^2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

<u>Goal</u>: Find a formula for $\vec{x}''$.



We know that $\vec{x}'' = k\vec{v}$, for some k.

We also know that $\vec{x}^{\perp} \perp \vec{v}$: $\vec{x}^{\perp}\cdot\vec{v} = 0$.

So, since $\vec{x} = \vec{x}'' + \vec{x}^{\perp}$, $\vec{x}^{\perp} = \vec{x} - \vec{x}''$:

$$\vec{x}^{\perp}\cdot\vec{v} = 0$$

$$(\vec{x} - \vec{x}'')\cdot\vec{v} = 0$$

$$\vec{x}\cdot\vec{v} - \vec{x}''\cdot\vec{v} = 0$$

$$\vec{x}\cdot\vec{v} - k\vec{v}\cdot\vec{v} = 0$$

$$\Rightarrow k = \frac{\vec{x}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}$$

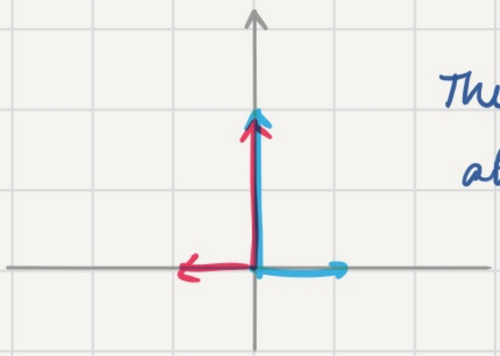
So, the projection is

$$\vec{x}'' = k\vec{v} = \left(\frac{\vec{x}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\right)\vec{v}.$$

Ex. $T(\vec{x}) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \vec{x}$

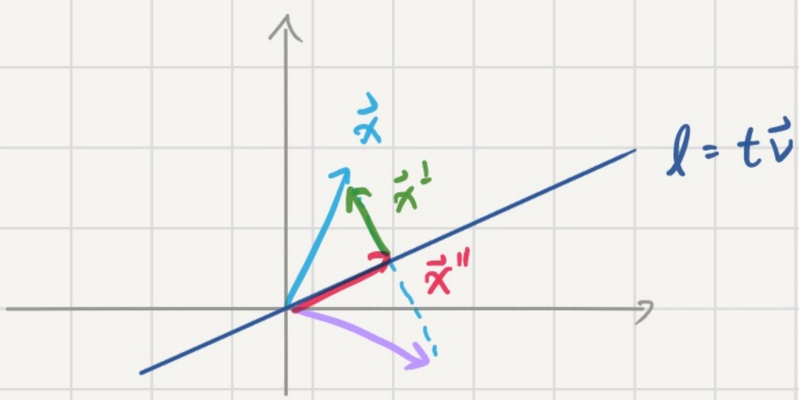
$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$



The "L" is reflected about the y-axis.

Reflection



The reflection of $\vec{x}$ about ℓ

$$\mathrm{ref}_\ell(\vec{x}) = \vec{x}^{\parallel} - \vec{x}^{\perp}.$$

This is just:

$$= \mathrm{proj}_\ell \vec{x} - (\vec{x} - \mathrm{proj}_\ell \vec{x})$$

$$= 2\,\mathrm{proj}_\ell \vec{x} - \vec{x}$$

A function $T: X \to Y$ is invertible if $T(\vec{x}) = \vec{y}$ has a unique solution $\vec{x} \in X$ for all $\vec{y} \in Y$:

$$T^{-1}(\vec{y}) = \vec{x}.$$

Note that

$$T^{-1}(T(\vec{x})) = \vec{x}, \qquad T(T^{-1}(\vec{y})) = \vec{y}.$$

Let A be an $n \times n$ matrix. Then, A is invertible if the linear transformation $T(\vec{x}) = A\vec{x}$ is invertible.

We let A^{-1} denote the standard matrix of the inverse transformation T^{-1}.

A matrix is invertible if and only if

$$T(\vec{x}) = A\vec{x} = \vec{y}$$

has a unique solution for all $\vec{y} \in \mathbb{R}^n$.

$$(\underset{n\times n}{A} \mid \vec{y}) \to \left(\begin{matrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{matrix} \middle| \begin{matrix} * \\ * \\ \vdots \\ * \end{matrix}\right)$$

<u>Theorem 3</u>. $A_{n\times n}$ is invertible if and only if $\mathrm{rref}(A) = I_n$.